reasoning bonds between steps, we flatten these matrices into vectors $\mathbf{p}$ and $\mathbf{q}$. The Pearson correlation coefficient is then calculated as:

$$r = \frac{\sum_{i=1}^{n}(p_i - \bar{p})(q_i - \bar{q})}{\sqrt{\sum_{i=1}^{n}(p_i - \bar{p})^2}\sqrt{\sum_{i=1}^{n}(q_i - \bar{q})^2}}, \tag{48}$$

where $\bar{p}$ and $\bar{q}$ are the mean values of vectors $\mathbf{p}$ and $\mathbf{q}$, respectively, and n is the number of elements in each vector. A high positive correlation (close to 1) indicates similar reasoning structures, while a low or negative correlation indicates dissimilar structures.

E Details about Synthetic Long CoT with Mole-Syn

This section describes the experimental settings used for synthesis, analysis, and reinforcement learning with Mole-Syn.

E.1 Supervised-Finetuning with Mole-Syn

To match the statistical properties of the synthetic chain-of-thought (CoT) to the behavioral patterns of stronger teacher models (Section C.2), we first estimate a reasoning-state transition distribution $p(s_{t+1}|s_t)$. Specifically, we analyze 20k distilled CoT rationales generated by teacher models (e.g., QwQ-32B and OpenAI-OSS-120B). Each reasoning step s_t is annotated by LLMs, and we use these annotations to compute an empirical transition matrix $\hat{P}$ (Figure 5). During synthesis, we initialize the process in an exploration state and then sample transitions between reasoning states according to. The prompts used for each reasoning state are listed below.

The prompt for the self-reflection state is as follows:

Prompt for Self-Reflection

Assume that you are a helpful assistant. You will receive a question and a previously reasoned rationale. If you can directly get the answer, please output the concise answer with \boxed{}. Otherwise, please reflect on the response and provide a self-reflection.
Here are some reasoning behavior definitions:

- normal operation — Straightforward, direct operations (e.g., arithmetic, factual recall, simple step-by-step logic) without introducing new logical nodes.
- deep reasoning — Multi-step causal, deductive, or analogical thinking that extends the reasoning chain by introducing new logical nodes or hidden assumptions.
- self-reflection — commenting on its own thought process (e.g., confidence, strategy, uncertainty, mistakes, or reconsideration of earlier reasoning) and tracing back to previous logical nodes.
- exploration — generating new possibilities, hypotheses, or questions, branching into alternative paths rather than following a single conclusion.

You should conduct self-reflection behavior now.

Please reflect on the response and provide a self-reflection.

The prompt for the self-exploration state is as follows:

Prompt for Self-Exploration

Assume that you are a helpful assistant. You will receive a question and a previously reasoned rationale. If you can directly get the answer, please output the concise answer with \boxed{}. Otherwise, please explore a novel reasoning path in the response.

Here are some reasoning behavior definitions:

Prompt for Self-Exploration (Continued)

- normal operation — Straightforward, direct operations (e.g., arithmetic, factual recall, simple step-by-step logic) without introducing new logical nodes.
- deep reasoning — Multi-step causal, deductive, or analogical thinking that extends the reasoning chain by introducing new logical nodes or hidden assumptions.
- self-reflection — commenting on its own thought process (e.g., confidence, strategy, uncertainty, mistakes, or reconsideration of earlier reasoning) and tracing back to previous logical nodes.
- exploration — generating new possibilities, hypotheses, or questions, branching into alternative paths rather than following a single conclusion.

You should conduct exploration behavior now.

Please explore a novel reasoning path in the response.

The prompt for the normal-operation state is as follows:

Prompt for Normal Operation

Assume that you are a helpful assistant. You will receive a question and a previously reasoned rationale. If you can directly get the answer, please output the concise answer with \boxed{}. Otherwise, please conduct normal operation on the response.

Here are some reasoning behavior definitions:

- normal operation — Straightforward, direct operations (e.g., arithmetic, factual recall, simple step-by-step logic) without introducing new logical nodes.
- deep reasoning — Multi-step causal, deductive, or analogical thinking that extends the reasoning chain by introducing new logical nodes or hidden assumptions.
- self-reflection — commenting on its own thought process (e.g., confidence, strategy, uncertainty, mistakes, or reconsideration of earlier reasoning) and tracing back to previous logical nodes.
- exploration — generating new possibilities, hypotheses, or questions, branching into alternative paths rather than following a single conclusion.

You should conduct normal operation behavior now.

Please conduct normal operation on the response.

The prompt for the deep-reasoning state is as follows:

Prompt for Deep Reasoning

Assume that you are a helpful assistant. You will receive a question and a previously reasoned rationale. If you can directly get the answer, please output the concise answer with \boxed{}. Otherwise, please further deepen the reasoning on the response.

Here are some reasoning behavior definitions:

- normal operation — Straightforward, direct operations (e.g., arithmetic, factual recall, simple step-by-step logic) without introducing new logical nodes.
- deep reasoning — Multi-step causal, deductive, or analogical thinking that extends the reasoning chain by introducing new logical nodes or hidden assumptions.
- self-reflection — commenting on its own thought process (e.g., confidence, strategy, uncertainty, mistakes, or reconsideration of earlier reasoning) and tracing back to previous logical nodes.
- exploration — generating new possibilities, hypotheses, or questions, branching into alternative paths rather than following a single conclusion.

You should conduct deep reasoning behavior now.

Please further deepen the reasoning on the response.

Model	GSM8K	MATH-500	AIME2024	AIME2025	AMC2023	OlympiadBench	AVG
LLaMA-3.1-8B-Base	7.58	3.20	0.00	0.00	4.22	1.19	2.70
+ 20K Qwen-Distill-Data	62.47	29.40	0.00	0.00	12.81	6.81	18.58
+ 20K OSS-Distill-Data	75.89	54.20	4.38	6.46	37.34	23.85	33.69
+ 20K QwQ-Distill-Data	64.53	32.20	2.92	0.42	16.72	8.89	20.95
+ 20K OSS-Mole-Syn	67.85	35.20	1.83	0.83	20.53	11.11	22.89
+ 20K QwQ-Mole-Syn	66.41	35.00	2.08	0.63	20.16	10.37	22.44
Llama-3.1-8B-Instruct	75.89	35.20	4.17	1.04	23.59	12.00	25.32
+ 20K Qwen-Distill-Data	76.50	39.80	4.38	1.04	25.63	19.70	27.84
+ 20K OSS-Distill-Data	79.00	60.80	10.83	7.71	47.03	30.22	39.27
+ 20K QwQ-Distill-Data	82.41	60.80	4.38	8.33	32.97	25.48	35.73
+ 20K OSS-Mole-Syn	83.24	51.80	4.79	1.04	32.50	21.04	32.40
+ 20K QwQ-Mole-Syn	84.31	50.20	5.21	1.67	32.34	20.00	32.29
Qwen-2.5-7B-Base	40.18	34.20	5.42	0.83	26.72	17.33	20.78
+ 20K Qwen-Distill-Data	68.69	39.80	4.38	1.04	25.63	19.70	26.54
+ 20K OSS-Distill-Data	84.99	68.40	6.04	8.13	46.25	27.70	40.25
+ 20K QwQ-Distill-Data	78.39	46.80	2.71	1.46	9.84	5.93	24.19
+ 20K QwQ-Mole-Syn	81.20	62.20	6.25	3.54	41.88	30.52	37.60
+ 20K OSS-Mole-Syn	83.17	63.80	5.83	1.67	41.56	29.33	37.56
Qwen-2.5-7B-Instruct	83.24	74.00	12.50	7.08	22.66	38.07	39.59
+ 20K Qwen-Distill-Data	84.31	63.40	6.46	3.13	31.72	29.78	36.46
+ 20K OSS-Distill-Data	82.31	74.60	25.42	19.38	54.34	37.63	48.94
+ 20K QwQ-Distill-Data	85.75	73.80	13.96	16.25	52.97	35.11	46.31
+ 20K QwQ-Mole-Syn	89.61	76.00	7.29	3.96	51.88	36.74	44.25
+ 20K OSS-Mole-Syn	88.02	77.80	8.13	5.00	52.81	37.48	44.87

Table 6 Comparison of strong reasoning LLM distillation versus Mole-Syn using a weak instructed LLM across six benchmarks.

Model	GSM8K	MATH-500	AIME2024	AIME2025	AMC2023	OlympiadBench	AVG
Llama-3.1-8B-Instruct	75.89	35.20	4.17	1.04	23.59	12.00	25.32
+ Mole-Syn-by-Llama-3.1-8B-Instruct	36.01	23.60	2.71	0.21	13.75	7.41	13.95
+ Mole-Syn-by-Llama-3.1-70B-Instruct	82.71	53.60	3.33	0.63	31.72	19.85	31.97
+ Mole-Syn-by-Qwen-7B	83.47	51.60	3.33	0.63	32.03	19.85	31.82
+ Mole-Syn-by-Qwen-32B	82.41	60.80	4.38	8.33	32.97	25.48	35.73

Table 7 The effectiveness analysis of Mole-Syn methods based on different instruct models.

Furthermore, to investigate the impact of distinct synthetic backbones, we employed Mole-Syn to distill data from various base models (Figure 7). Models with limited self-reflection or exploration capabilities, such as Llama-3.1-8B-Instruct, struggled to synthesize effective reasoning data. Conversely, combining the Qwen series with Llama-70B yielded robust results across most domains, with the notable exception of tasks requiring deep reasoning. For such tasks (e.g., AIME), performance depended heavily on the model's intrinsic reasoning depth, following the hierarchy: Qwen-32B outperforms both Qwen-7B and Llama-70B.

E.2 Reinforcement Learning with Mole-Syn Initialization

We adopt DAPO [40] as the reinforcement-learning (RL) framework to fine-tune Llama-3.1-8B-Instruct models initialized from different checkpoints. To isolate the effect of initialization, we hold the RL tasks, reward functions, and all other training hyperparameters constant across experiments. Each model is trained for 1000 steps with a learning rate of 1×10^{-5}, a batch size of 16, a sampling size of 16, and a maximum sequence length of 16384 tokens. We set the clipping parameters to clip-low = 0.2 and clip-high = 0.68. We use MATH [26] and AIME 1989–2023 [41] as training data.

We evaluate two RL initialization strategies: (1) **QwQ-Distill Data + RL:** RL fine-tuning starting from Llama-3.1-8B-Instruct further tuned on QwQ-generated long chain-of-thought data. (2) **QwQ-Graph-Syn + RL:** RL fine-tuning starting from our Mole-Syn-SFT model. All runs use the same RL tasks, reward functions, and hyperparameters; therefore, performance differences can be attributed to the initialization checkpoint (i.e.,